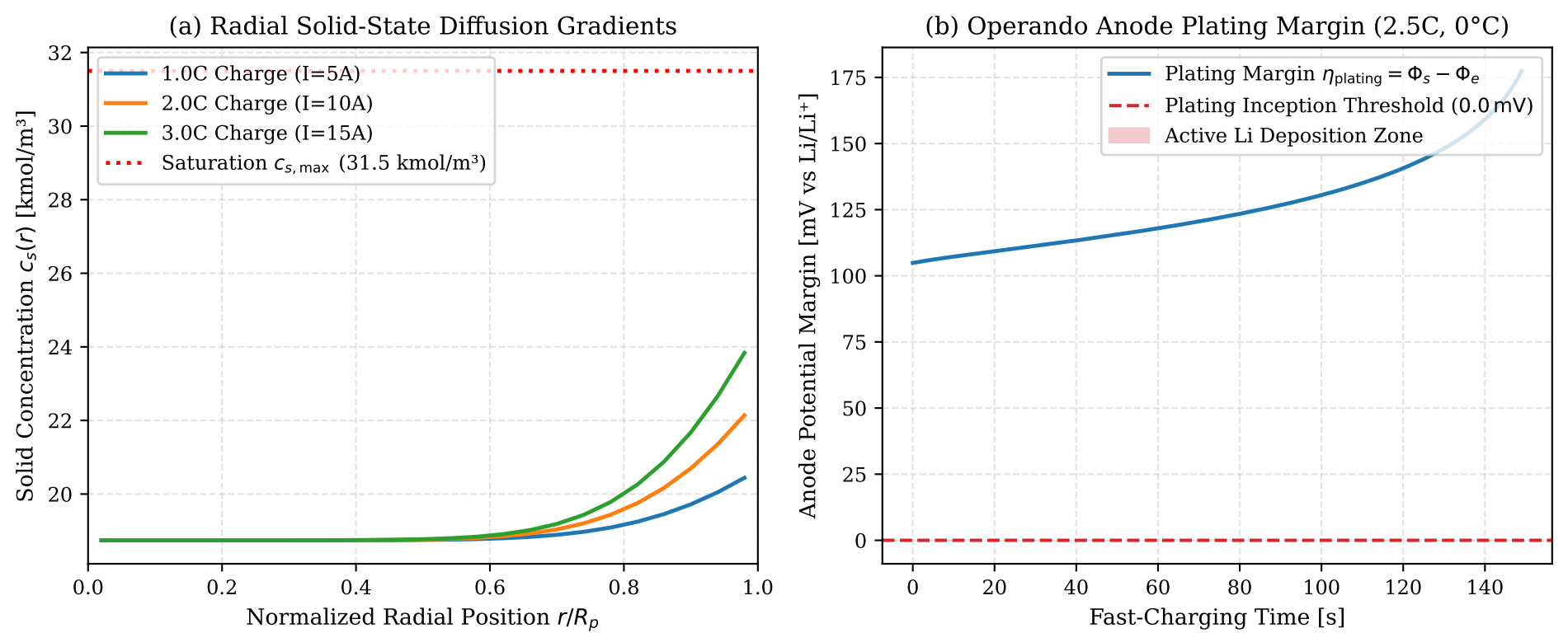
(a) Radial Solid-State Diffusion Gradients
1.0C Charge (I=5A)
2.0C Charge (I=10A)
3.0C Charge (I=15A)
Saturation $c_{s,\max}$ (31.5 kmol/m³)
Solid Concentration $c_s(r)$ [kmol/m³]
Normalized Radial Position r/R_p
32
30
28
26
24
22
20
0.0
0.2
0.4
0.6
0.8
1.0
(b) Operando Anode Plating Margin (2.5C, 0°C)
Plating Margin $\eta_{\text{plating}} = \Phi_s - \Phi_e$
Plating Inception Threshold (0.0 mV)
Active Li Deposition Zone
Anode Potential Margin [mV vs Li/Li⁺]
Fast-Charging Time [s]
175
150
125
100
75
50
25
0
0
20
40
60
80
100
120
140